Statement of Purpose

Tanzeem Tahmeed Reza, Ph.D Applicant, Fall 2026 tanzeemreza19@gmail.com

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Growing Interest: I have always been fascinated by films, and a science fiction movie first sparked my interest in neural signal processing. I remember from one of them, a system reconstructed visual scenes directly from a person's thoughts, prompting me to ask: ***can neural signals meaningfully represent what humans perceive, imagine, or intend to express?*** This curiosity evolved into a research-driven pursuit, focusing on neural signal decoding for assistive communication. My current vision centers on modeling neural representations of speech and motor control to develop technologies that support individuals with speech motor disorders.

Research Journey: My early research focused on **neural signal generation and analysis.** For my undergraduate thesis, I developed three independent Wasserstein GAN–based pipelines to synthesize EEG signals evoked by digit-based visual stimuli, addressing the challenge of limited neural datasets and shifting my perspective from EEG as raw data to a meaningful representation of cognitive processes. Building on this, I conducted and published a study on **emotion detection using EEG** in collaboration with CUET's Biomedical Engineering Department, developing robust preprocessing and classification pipelines in MATLAB and MNE-Python. I co-led a study on sleep stage classification using EEG and XAI (manuscript under review in a Q1 journal), and published papers on **Parkinson's detection via handwriting signals** and **breast cancer image classification.** These experiences strengthened my ability to design end-to-end machine learning pipelines across diverse biomedical modalities, preparing me for the interdisciplinary demands of neural speech decoding and assistive speech technology research.

Current Research Interest: I am pursuing a Ph.D. to receive rigorous training in hypothesis-driven research, experimental design, and translational neuroscience that integrates speech science with advanced computational methods. At this stage, I aim to move

beyond algorithmic performance alone and develop a deeper understanding of speech motor control, neural mechanisms underlying speech production, and clinically grounded research questions. Doctoral training in Speech, Language, and Hearing Sciences (SLHS) will allow me to contextualize my technical expertise within the theoretical and clinical frameworks necessary for impactful assistive communication research. The **Ph.D. program in SLHS** at The **University of Texas at Austin** can provide me with an ideal environment for this goal due to its strong interdisciplinary culture and close integration with neuroscience and clinical research through Moody College of Communication and Dell Medical School.

In particular, I am highly interested in the **Speech Disorders and Technology Lab (SDTL)** led by **Dr. Jun Wang**. I have carefully studied Dr. Wang's work, including research on direct speech synthesis from neuromagnetic signals and automatic detection of ALS from single-trial MEG data during speech tasks. The lab's focus on motor speech disorders, neural speech decoding, and assistive speech technologies closely aligns with my research interests and background. I am also drawn to **Dr. Rosemary Lester-Smith's Voice Lab**, where my signal processing and machine learning expertise could contribute to clinically feasible assessments for neurological voice disorders and translational BCI research.

Future Career Goal: My long-term goal is to pursue an **academic research career** focused on developing clinically relevant neural speech technologies that improve communication outcomes for individuals with speech motor disorders. Through doctoral training at UT Austin, I aim to contribute original research at the intersection of speech science, neuroscience, and machine learning, while gaining the clinical and theoretical grounding necessary for translational impact. I am confident that the SLHS Ph.D. program and the research environment within the Speech Disorders and Technology Lab will provide the ideal foundation for achieving these goals.